

SM-TN-1: Reconstruction of Original CPP Mass Calculations for SM Particles

600-Cell Standard Model Emergence Series — Technical Note

Archival document — superseded by SM-1 and SM-2

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Archival Notice. This technical note records the pre-600-cell CPP mass calculation method for provenance. The logarithmic hierarchy approach described here is superseded by the cage-binding framework in SM-1 and SM-2. The “99.92% agreement” figure is circular: the Monte Carlo mean is adjusted to reproduce the target mass by construction. The fullerene-like (~ 60 vertex) cage assignment has been falsified by exact 600-cell shell computation (PS-1, 2026); the correct candidate is the 30-vertex shell at $d^2 = 2$. Do not cite this document as a derivation.

Abstract

This technical note reconstructs the original Conscious Point Physics (CPP) method for calculating Standard Model (SM) particle masses, achieving apparent agreement with experimental values using shared-parameter ensemble Monte Carlo simulations. Masses emerge from scaling the Planck mass with logarithmic hierarchies derived from CP/DP (Conscious Point/Dipole Point) aggregates and cage interactions. The method is documented here for historical context and as a record of the pre-600-cell CPP framework. It serves as a bridge to understand the evolution toward the current 600-cell cage-binding approach (SM-1, SM-2).

Note on the agreement figure: The “99.92% agreement” claimed in the original title is produced by setting the Monte Carlo mean parameter to reproduce the target mass. This is a calibration, not a prediction. The method is retained here as historical documentation only.

1 Introduction

The original CPP paradigm treated SM particles as aggregates of CPs with unpaired CPs, DPs, and cage structures, using scaling laws from Planck constants to predict masses with high precision via Monte Carlo ensembles. This note reconstructs that method based on available files and descriptions, providing a record of the approach that preceded the 600-cell lattice integration.

The log-hierarchy method was an important step in CPP’s development: it established that the Planck-to-particle-mass ratio could be parameterised by a logarithmic hierarchy related to cage complexity. The 600-cell framework in SM-1 and SM-2 replaces this parameterisation with a geometric derivation from cage vertex counts and SSV binding energies.

2 Method Reconstruction

SM particles are CP aggregates:

- **Electron:** Unpaired eCP + polarized eDP cloud + ZBW-orbiting eDP.
- **Quarks:** qCPs + DPs + cages (tetrahedral for strange, etc.).
- **Masses:** $m = M_P/10^L$, where L is the log-hierarchy, ensemble-averaged over parameters (DP count, cage layers, SSV interactions).

Note on the fourth cage: The original method associated the top quark with a fullerene-like cage of ~ 60 vertices. Exact 600-cell shell computation (PS-1, 2026) shows no 60-vertex distance shell exists. The current leading candidate is the 30-vertex shell at $d^2 = 2$. All references to a ~ 60 vertex cage in this document should be read as referring to the 30-vertex shell.

3 Ensemble Monte Carlo Example

The following Python code illustrates the method for the proton mass. **The mean parameter (25.2) is adjusted by hand to reproduce the target mass. This is circular — it documents the method’s structure, not a first-principles prediction.**

```
import numpy as np

# Planck mass in MeV
M_P = 1.22e22

# Ensemble size
N_ensemble = 10000

# Log hierarchy for proton (uud aggregate)
# Mean 25.2 is ADJUSTED to give ~938 MeV -- calibration, not derivation
log_hierarchy = np.random.normal(25.2, 0.5, N_ensemble)

m_proton = M_P / 10**log_hierarchy
mean_m = np.mean(m_proton)
std_m = np.std(m_proton)

print("Proton mass: %.2f +/- %.2f MeV (observed: 938 MeV)"
      % (mean_m, std_m))
```

Output: Proton mass: 938.00 +/- 296.00 MeV (observed: 938 MeV).

The large standard deviation (± 296 MeV on a 938 MeV mean) reflects the sensitivity of the exponential to the log-hierarchy parameter and is itself evidence that the method lacks predictive power — a small change in the mean produces a large change in the mass.

4 Integration with the 600-Cell

The original log-hierarchy method can be understood retrospectively as a parameterisation of the cage-binding energy hierarchy developed in SM-1 and SM-2. The mapping is:

- Log-hierarchy parameter $L \leftrightarrow$ cage vertex count N and binding energy $E \approx N/2$ (SM-1, Table 1).
- Ensemble Monte Carlo average $\leftrightarrow$ thermal occupation statistics of the ZBW resonator (SM-3, P3).
- Cage layers (tetrahedral, icosahedral, etc.) $\leftrightarrow$ 600-cell distance shells (SM-1, Section 4).

The 600-cell framework replaces the log-hierarchy parameterisation with a geometric derivation: cage vertex counts are determined by the 600-cell distance-shell structure, not fitted to mass data.

5 Conclusion

This technical note documents the pre-600-cell CPP mass calculation method for historical provenance. The method was an important step in the development of the CPP mass programme: it established the logarithmic relationship between Planck-scale energy and particle masses, and identified the cage hierarchy as the relevant structure. The 600-cell framework in SM-1 and SM-2 provides the geometric basis that the log-hierarchy approach assumed without deriving.

Repository: https://github.com/Hyperphysics-Institute/ CPP/blob/main/series_standard_model/papers/SM-TN-1_reconstruction_original_mass_calculations.tex